

PROJECTIVE POLYNOMIAL OBSERVATIONS: NORMALIZATION, SPECTRUM, AND SINGULAR INFERENCE

QIAN QI

ABSTRACT. We study matrix-valued probability curves obtained by normalizing a simultaneously diagonalizable polynomial through two unknown stochastic channels. A linear operator formed from the observations characterizes recovery of the monic normalization. With full-rank channels, at $2d + 1$ clocks its nullity is exactly the degree of the common divisor of the component polynomials; this clock count is sharp in the worst case, even with fixed positive channels. Its smallest singular value and the product of the channel singular values enter an additive, one-sided spectral modulus. The modulus allows arbitrary competitors in the closed model and collisions between components. In degree one we replace latent-factor optimization by a canonical resolvent-residue invariant, identify its zero set, and prove its semicontinuity and a stronger Lipschitz inverse. A direct least-squares, singular-subspace and eigenvalue estimator attains this inverse order. Honest confidence sets and matching minimax orders follow on observable signal classes; two positive least-favourable families remain least favourable under adaptive clock selection.

1. INTRODUCTION

Normalization removes a scalar amplitude from a matrix observation. It can also remove a polynomial factor, and with it part of a latent spectrum. The inverse problem therefore has two stages: recover a normalization from projective data, and recover the spectrum of the resulting matrix polynomial. Treating these stages as two unrelated condition numbers can introduce a loss which the observation map does not possess. The normalization error remains diagonal in the true channel coordinates; it need not be amplified a second time by the channel inverses.

We give a finite-clock formulation of this principle. The normalization is characterized by an observable linear residual operator. For polynomial degree d , its kernel has an exact algebraic description once $2d + 1$ clocks are available. For an arbitrary design, its rank is the identification criterion, and its least singular value is the normalization condition. This separates a worst-case clock count from the information actually present in a particular curve.

The affine experiment admits a further reduction. Its normalized curve has the form $A + B/(T - h)$. After recovering h , the residues of the inverse reduced pencil give an intrinsic condition, including at repeated eigenvalues. The condition is a finite observable construction, not a maximum over latent factorizations. It has a specified zero set and a sharp direction of semicontinuity. The spectral target remains Lipschitz through partial action collisions, although the factors and the residue partition may change. Two different boundaries—channel rank loss and collapse of the entire projective curve—give sharp statistical orders.

1.1. The polynomial experiment. Fix $d \geq 1$, $2 \leq k \leq \min(m_1, m_2)$, an interval $[L, D]$, distinct clocks $D < T_1 < \dots < T_\ell$, with $\ell \geq d + 1$, and $0 < \alpha_* < 1/k$. The columns of $U \in \mathbb{R}^{m_1 \times k}$

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and $V \in \mathbb{R}^{m_2 \times k}$ are probability vectors. Let $\alpha_b \geq \alpha_*$ and $\sum_b \alpha_b = 1$. Put

$$\begin{aligned} f_b(z) &= \prod_{a=1}^d (z - \lambda_{ba}), \quad \lambda_{ba} \in [L, D], & q(z) &= \sum_b \alpha_b f_b(z), \\ K(z) &= U \operatorname{diag}(\alpha_b f_b(z)) V^\top, & P(T) &= \frac{K(T)}{q(T)}. \end{aligned} \quad (1.1)$$

The parameter set $\mathcal{K}_d$ is compact; singular channels and repeated roots are included. The polynomials q and f_b are monic of degree d . All $q(T_j)$ and component probabilities $p_b(T_j) = \alpha_b f_b(T_j)/q(T_j)$ have uniform positive bounds. An arbitrary scalar detector multiplying $K(T)$ disappears on normalization. Component-dependent detectors are not part of this experiment.

The target Λ is the unordered multiset of the kd roots of all f_b , counted with multiplicity. We do not assert recovery of a partition of these roots into components. For two multisets of the same cardinality write

$$d_\infty(\Lambda, \Lambda') = \min_\pi \max_a |\lambda_a - \lambda'_{\pi(a)}|, \quad \delta = \max_j \|P(T_j) - P'(T_j)\|_F.$$

For real multisets this is the maximum distance between their increasing orders. All constants in this paper may depend on $d, k, m_1, m_2, \alpha_*, L, D$ and the fixed clocks, but not on action gaps, unless a gap is explicitly specified. There is no assertion uniform in growing dimension or a vanishing weight floor.

Let $W = (T_j^r)_{j=1, \dots, \ell; r=0, \dots, d}$ and let Ω be the orthogonal projector onto $\operatorname{col}(W)^\perp$. Write $q(z) = z^d + \sum_{r < d} a_r z^r$. For $c \in \mathbb{R}^d$ set $r_c(z) = \sum_{r < d} c_r z^r$ and define

$$\begin{aligned} \mathcal{N}_P &= (\Omega \otimes I)(r_c(T_j) \operatorname{vec} P(T_j))_{j=1}^\ell, & \tau(P) &= \sigma_{\min}(\mathcal{N}_P), \\ b_P &= -(\Omega \otimes I)(T_j^d \operatorname{vec} P(T_j))_{j=1}^\ell, & \kappa &= \sigma_k(U) \sigma_k(V). \end{aligned} \quad (1.2)$$

Here I has size $m_1 m_2$, and the singular value uses Euclidean coefficient norm and the product Frobenius norm. Thus τ is a specified observable quantity, not a coordinate-free number without a choice of norms. Changing an orthonormal basis of the residual space does not change it. The normalization equation is $\mathcal{N}_P a = b_P$.

Theorem 1.1 (Normalization, clock design, and spectral recovery). *Suppose the true channels have full column rank. If $\ell \geq 2d + 1$ and $g = \gcd(f_1, \dots, f_k)$ is monic, then*

$$\ker \mathcal{N}_P = \{(q/g)s : \deg s < \deg g\}, \quad \dim \ker \mathcal{N}_P = \deg g, \quad (1.3)$$

where polynomials of degree less than d are identified with their coefficient vectors. In particular $\tau(P) > 0$ exactly when the component polynomials are relatively prime as a collection.

For any design with $\ell \geq d + 1$ and $\tau(P) > 0$, every competitor in $\mathcal{K}_d$ satisfies

$$\|a - a'\|_2 \leq C \min\{1, \delta/\tau(P)\}, \quad (1.4)$$

$$d_\infty(\Lambda, \Lambda') \leq C \min\{1, [\delta(\kappa^{-1} + \tau(P)^{-1})]^{1/d}\}. \quad (1.5)$$

If the roots within each true f_b are simple and separated by at least $\nu > 0$, the exponent in (1.5) is 1, with a constant also depending on ν . Roots belonging to different components may coincide. No rank or root-separation assumption is made on the competitor.

Conversely, at an interior parameter with simple roots within each component, $\tau(P) = 0$ gives a nonconstant local exact fibre with the same channels and a changing root multiset. The count $2d + 1$ is sharp in the worst case for every fixed admissible k : there are relatively prime, interior examples for which $2d$ clocks do not identify the multiset. Richer curves can satisfy the rank criterion at fewer clocks.

In degree one write $f_b(z) = z - \xi_b$ and

$$h = \sum_b \alpha_b \xi_b, \quad A = U \operatorname{diag}(\alpha) V^\top, \quad C = U \operatorname{diag}(\alpha_b \xi_b) V^\top, \quad B = hA - C. \quad (1.6)$$

Then $P(T) = A + B/(T - h)$ and $K(z) = zA - C$. At three clocks $\tau(P) \asymp \gamma(P) := \|B\|_F$. Consequently (1.5) recovers the gap-free affine inverse with $\vartheta = \kappa\gamma/(\kappa + \gamma)$, set to zero on either boundary. Section 3 strengthens this to an observable condition $\eta(P)$ constructed from inverse-pencil residues. It satisfies $\eta(P) \geq c\vartheta_*(P)$, where

$$\vartheta_*(P) = \max\{\vartheta(\omega) : \mathcal{P}(\omega) = P\}. \quad (1.7)$$

Theorem 3.2 gives a one-sided Lipschitz bound with η in place of ϑ_* . Theorem 4.1 gives a direct spectral estimator, and Theorem 5.4 proves matching minimax orders on both signal classes. These are worst-case class orders, with pointwise lower bounds on two explicit families; they are not a claim that every anisotropic singular point has the same local minimax geometry.

1.2. Relation to previous work. The equation $\mathcal{N}_P a = b_P$ is a vector rational-interpolation condition. Rational interpolation and its solution spaces have a substantial history; see Van Barel–Bultheel [15]. We do not claim that linearizing rational interpolation is new. The content here is the identification of its defect with the common latent polynomial factor, its realization by positive fixed-channel fibres, and its combination with a one-sided spectral modulus in observation norm.

Nor is multiset perturbation of a regular pencil new. Bauer–Fike [2] provides the basic exclusion argument. Chu’s generalized pencil theorem [6] already treats eigenvalue clusters and relates conditioning to regularity and eigenvector deficiency; his matrix-polynomial extension [7] includes multiple eigenvalues of monic and regular polynomials. Our root-counting lemma is an explicit specialization of this resolvent-and-homotopy mechanism, not a claim to have discovered perturbation through repeated eigenvalues. The additional step is recovering an unknown scalar polynomial from normalized observations while retaining the diagonal form of its error.

Higham–Mackey–Tisseur [10] compare the conditioning of a matrix polynomial with that of its linearizations. In particular their simple-eigenvalue formula involves the derivative of the polynomial; it is not a uniform estimate through all component-root collisions. Tisseur–Higham [14] develop structured polynomial pseudospectra, and Su–Bai [13] study linearization of rational eigenvalue problems. Those results start with a specified polynomial or rational matrix function and a coefficient perturbation model. The present normalization rank, positive exact fibres, and additive conversion from categorical observation error are separate assertions. Our residue invariant uses the classical resolvent; the new bound also uses the exact projective pole-shift fibre before applying it.

Observable latent diagonalization and factor identification are classical [1, 4]. Robust tensor uniqueness, including quantitative Kruskal-type results, is treated by Bhaskara–Charikar–Vijayaraghavan [3]. Here the target is a multiset, not a stable naming of individual rank-one factors; repeated actions can destroy the latter without destroying the former. Li–Lee–Bresler [12] give a general nuisance-transformation perspective on bilinear identification. The passage from inverse moduli to inference is classical [8, 5, 9], as is the occurrence of singular rates in finite mixtures [11]. The target-dependent distinction between partial spectral collisions and complete projective collapse is established below for the stated controlled categorical experiment.

2. NORMALIZATION AND THE FINITE-CLOCK FIBRE

The normalizing polynomial is constrained by the requirement that $q(T_j)P(T_j)$ interpolate a matrix polynomial of degree at most d . The operator in (1.2) records precisely this constraint. We first determine its nullspace.

Lemma 2.1 (The normalization defect). *Under the rank assumption of Theorem 1.1, the greatest common divisor of all entries of K is $g = \gcd(f_1, \dots, f_k)$. If $\ell \geq 2d + 1$, then (1.3) holds.*

Proof. Every entry of K is a constant linear combination of the f_b , so g divides those entries. Conversely, multiplication by the constant left inverse $U^\dagger$ and right inverse $(V^\dagger)^\top$ recovers $\text{diag}(\alpha_b f_b)$. Every common divisor of the entries therefore divides every f_b .

Let r have degree less than d and belong to $\ker \mathcal{N}_P$. There is a matrix polynomial H of degree at most d such that $H(T_j) = r(T_j)P(T_j)$. Each entry of $qH - rK$ has degree at most $2d$ and vanishes at $2d + 1$ distinct clocks. Hence $qH = rK$ identically. Write $K = g\tilde{K}$ and $q = g\tilde{q}$. The entries of $\tilde{K}$ have greatest common divisor one. Their Bezout identity implies that $\tilde{q}$ divides r . Thus $r = \tilde{q}s$, where $\deg s < \deg g$. Conversely, for every such s the polynomial $H = \tilde{K}s$ has degree at most $d - 1$ and satisfies $qH = rK$. This proves both assertions, including the case $\deg g = d$. $\square$

The defect is an actual normalization ambiguity. Cancelling g changes the visible degree from d to $d - \deg g$; the zeros of the cancelled factor cannot be read from the normalized curve. At a shorter design, aliasing can produce a normalization defect even when $g = 1$.

Lemma 2.2 (Positive local fibres). *Suppose $\ell \geq d + 1$, the true channels have full column rank, $\alpha_b > \alpha_*$, and all roots of each f_b are simple and lie in (L, D) . If $0 \neq r \in \ker \mathcal{N}_P$, there is a smooth curve of parameters with the same U, V , unchanged observations at every sampled clock, and a nonconstant root multiset.*

Proof. Let H be the degree-at-most- d interpolant of $r(T_j)P(T_j)$. All its values lie in the space $\{U \text{diag}(x_b)V^\top : x \in \mathbb{R}^k\}$. Interpolation at the first $d + 1$ nodes shows that $H(z) = U \text{diag}(h_b(z))V^\top$ with $\deg h_b \leq d$. Since the sum of the entries of $P(T_j)$ is one, the polynomial $\sum_b h_b - r$ vanishes at $d + 1$ nodes and is identically zero.

For small real u , put

$$q_u = q + ur, \quad K_u = K + uH, \quad w_{b,u} = \alpha_b f_b + u h_b.$$

Let $\alpha_{b,u}$ be the leading coefficient of $w_{b,u}$ and set $f_{b,u} = w_{b,u}/\alpha_{b,u}$. Then $\sum_b \alpha_{b,u} = 1$ and $q_u = \sum_b \alpha_{b,u} f_{b,u}$. The strict weight inequalities persist. Real simple roots in an open interval persist under sufficiently small real coefficient perturbations: choose disjoint brackets about the roots with opposite signs at their endpoints. Each bracket retains a root, accounting for all d roots. These roots are smooth by the implicit function theorem. Moreover $K_u(T_j) = q_u(T_j)P(T_j)$, as required.

If the root multiset were unchanged for sufficiently small nonzero u , each component root, being close to its original value, would have to equal that value. Thus $f_{b,u} = f_b$ for every b . Applying the true channel inverses to the equality of observations then gives $\alpha_{b,u}/\alpha_b = q_u(T_j)/q(T_j)$ for every b, j . Summing the weights makes this common ratio one. Consequently r vanishes at $d + 1$ nodes, which contradicts $r \neq 0$ and $\deg r < d$. $\square$

Proposition 2.3 (Sharp worst-case clock count). *For every admissible fixed k, m_1, m_2 , there are positive full-rank channels and interior, relatively prime component polynomials for which any specified $2d$ distinct clocks admit the fibre of Lemma 2.2.*

Proof. First take $k = 2$ and write $P(T) = E_2 + (E_1 - E_2)w(T)$, where $E_b = U_{:,b}V_{:,b}^\top$ and $w = \alpha_1 f_1/q$. Since $\Omega(r(T_j))_j = 0$ for $\deg r < d$,

$$\mathcal{N}_{Pr} = [\Omega(r(T_j)w(T_j))_j] \otimes \text{vec}(E_1 - E_2).$$

At $2d$ nodes the residual scalar space has dimension $d - 1$. Thus the map from the d coefficients of r has nonzero kernel, including when f_1, f_2 are relatively prime.

For arbitrary k , choose two close, relatively prime monic polynomials f, g with simple roots in (L, D) . They can be chosen so that every $(1 - s)f + sg$, $0 \leq s \leq 1$, has simple roots in that interval, by the bracket argument in the preceding lemma. Choose $s_b \in [0, 1]$, including both endpoints, and set $f_b = (1 - s_b)f + s_bg$. The collection is relatively prime. Its matrix numerator belongs to the span of two fixed matrices, and normalization restricts it to an affine line, since every observed matrix has entry sum one. The same residual-dimension argument applies. Take $\alpha_b = 1/k$ and any strictly positive full-rank stochastic channels. Lemma 2.2 finishes the proof. $\square$

2.1. Root matching without cross-component gaps. For a polynomial matrix $F(z) = \sum_{r=0}^d F_r z^r$, write $\|F\|_{\text{coef}} = \sum_r \|F_r\|_{\text{op}}$. The next lemma states explicitly the root-counting step used below.

Lemma 2.4 (Diagonal polynomial perturbations). *Let $D_f(z) = \text{diag}(\alpha_b f_b(z))$ be as in (1.1). If $\|E\|_{\text{coef}} \leq \varepsilon$ is sufficiently small, then $\det(D_f + E)$ has degree kd and its root multiset Z_E satisfies*

$$d_\infty(\Lambda, Z_E) \leq C\varepsilon^{1/d}.$$

The roots in Z_E may be complex. If the roots within each f_b are simple and ν -separated, the bound is $C_\nu \varepsilon$. No separation between roots of different components is required.

Proof. The leading coefficient of $D_f + sE$ is invertible for $0 \leq s \leq 1$, provided $\varepsilon < \alpha_*/2$. For $|z|$ larger than a fixed constant, $|f_b(z)| \geq c|z|^d$, whereas $\|E(z)\| \leq C\varepsilon|z|^d$. A Neumann series therefore excludes zeros outside a fixed disk, uniformly in s . Inside that disk, if $\rho = \text{dist}(z, \Lambda)$, then

$$\|D_f(z)^{-1}\| \leq \alpha_*^{-1} \rho^{-d}, \quad \|E(z)\| \leq C\varepsilon.$$

Thus every zero along the homotopy lies in the union of disks of radius $r = (C_1 \varepsilon)^{1/d}$ about the true roots, with C_1 fixed large enough to make the exclusion strict outside these disks.

For completeness, fix any $r' > r$ which is not one of the finitely many radii at which two of the disks first touch. Distinct components of the union of the closed r' -disks have positive distance from one another. Each can be surrounded by finitely many piecewise smooth contours contained in the complement of the closed r -disks, enclosing that component and no other. Slight outer perturbations of the boundary of its disk union give such contours; holes may be filled, since the zeros already lie in the r -disk union. The determinant is nonzero on the contours for the whole homotopy. The argument principle therefore keeps the number of enclosed roots, counted with algebraic multiplicity, equal to its value at $s = 0$.

If a component contains n centres with multiplicity, a chain of its disks joins any two distinct centres in at most $n - 1$ steps. Their distance is at most $2(n - 1)r'$. Every root enclosed in that component is within r of one of its centres. Since there are exactly n roots, any bijection inside the component has distance at most $(2n - 1)r'$. Let $r' \downarrow r$ through nonexceptional radii. There are only finitely many permutations, so one can take a subsequence with a fixed matching. This proves the asserted estimate, including touching components and repeated centres. For real spectra, uncrossing inverted pairs proves the equivalent ordered matching.

Under the within-component separation assumption, near a root of f_b all its other factors have magnitude bounded below in terms of ν . Away from those neighbourhoods $|f_b|$ is uniformly positive. On the fixed disk this gives $|f_b(z)| \geq c_\nu \min\{1, \text{dist}(z, Z(f_b))\}$. The same argument then uses radius $C_\nu \varepsilon$. $\square$

Proof of Theorem 1.1. The kernel and fibre assertions have been proved. For a competitor, $\mathcal{N}_{P'} a' = b_{P'}$. Projecting its cleared observations gives

$$\mathcal{N}_P(a' - a) = (\Omega \otimes I)(q'(T_j) \text{vec}(P(T_j) - P'(T_j)))_j. \quad (2.1)$$

Uniform coefficient and clock-value bounds for q' imply $\|a' - a\| \leq C\delta/\tau$. Compactness supplies the cap in (1.4).

Let $L_U = U^\dagger$, $R_V = (V^\dagger)^\top$, so that $\|L_U\|\|R_V\| = \kappa^{-1}$. Set $M(z) = L_U K'(z) R_V$. At the first $d + 1$ clocks,

$$\begin{aligned} M(T_j) - \text{diag}(\alpha_b f_b(T_j)) &= q'(T_j) L_U (P'(T_j) - P(T_j)) R_V \\ &\quad + (q'(T_j) - q(T_j)) \text{diag}(p_b(T_j)). \end{aligned} \quad (2.2)$$

The second term is diagonal. Interpolation at this fixed design gives

$$\|M - D_f\|_{\text{coef}} \leq C\delta(\kappa^{-1} + \tau^{-1}) =: C\epsilon.$$

In particular the normalization error has not acquired a factor κ^{-1} . For small ϵ , the leading coefficient of M is invertible. It is also $(L_U U') \text{diag}(\alpha') (V'^\top R_V)$, so both square outer factors must be invertible. Therefore

$$\det M(z) = \det(L_U U') \det(V'^\top R_V) \prod_b \alpha'_b f'_b(z).$$

Its roots are exactly the competitor multiset, even though no initial rank assumption was made on that competitor. Lemma 2.4 proves both small-error bounds. For larger ϵ , all target roots belong to $[L, D]$, which proves the capped estimates. $\square$

Remark 2.5. The exponent $1/d$ is a uniform upper bound, not a claim of an optimal exponent at every real-rooted singularity. The linear result requires separation within a component polynomial, not separation of the aggregate spectrum. In degree one this additional assumption is empty. For $d > 1$, the kernel identity identifies common-factor cancellation, whereas the root modulus also records multiple roots within a scalar component. These are different mechanisms.

3. THE AFFINE QUOTIENT AND ITS OBSERVABLE RESIDUE CONDITION

Throughout this section $d = 1$ and $\ell = 3$. The parameters are as in (1.6). Write $P_j = P(T_j)$ and $\gamma = \|B\|_F$.

Lemma 3.1 (The third clock and the pole-shift fibre). *If $\gamma > 0$, put $X = P_1 - P_2$, $Y = P_2 - P_3$, and $a_0 = (T_2 - T_1)/(T_3 - T_2)$. Then*

$$r := \frac{\langle X, Y \rangle_F}{\|Y\|_F^2} = a_0 \frac{T_3 - h}{T_1 - h}, \quad h = \frac{rT_1 - a_0T_3}{r - a_0}, \quad \|Y\|_F \asymp \gamma. \quad (3.1)$$

The map from $h \in [L, D]$ to r is uniformly bi-Lipschitz. If two model observations are at distance $\delta \leq c\gamma$, then $|h - h'| \leq C\delta/\gamma$. Also $\tau(P) \asymp \gamma$.

Let $H_0 = (T_1 - h)(T_2 - h)$. For sufficiently small real u , define

$$\begin{aligned} a_x(u) &= 1 + u(h - x)/H_0, \\ x_u &= \frac{x + u\{1 + (h - x)(T_1 + T_2 - h)/H_0\}}{a_x(u)}, \\ \alpha_{b,u} &= \alpha_b a_{\xi_b}(u), \quad h_u = h + u. \end{aligned} \quad (3.2)$$

With the same channels, these parameters give exactly the same two observations at T_1, T_2 , whenever they remain admissible. At interior parameters they give a nonconstant exact fibre, since

$$\left. \frac{dx_u}{du} \right|_{u=0} = \frac{(T_1 - x)(T_2 - x)}{H_0} > 0. \quad (3.3)$$

Proof. Subtract $P(T) = A + B/(T - h)$ at consecutive clocks. This gives (3.1); the derivative of $r(h)$ is $a_0(T_3 - T_1)/(T_1 - h)^2$, bounded above and below by positive constants. At error $\delta \leq c\gamma$, both difference norms are bounded above by $C\gamma$ and the perturbed denominator norm is bounded below by $c\gamma$. Expanding the ratio gives error $C\delta/\gamma$, hence the pole estimate.

For $d = 1$, $\mathcal{N}_P$ has one column. Since Ω annihilates constant and affine clock vectors,

$$\tau(P) = \|B\|_F \left\| \Omega((T_j - h)^{-1})_{j=1}^3 \right\|_2.$$

The last norm is continuous and strictly positive for $h \in [L, D]$: if a degree-one polynomial agreed with $(T - h)^{-1}$ at three distinct points, multiplying by $T - h$ would give an impossible polynomial identity. Compactness proves the comparison.

The affine interpolant of $p_b(T)$ at T_1, T_2 is

$$Q_b(z) = \alpha_b \left[1 + (h - \xi_b) \frac{T_1 + T_2 - h - z}{H_0} \right].$$

The polynomial $\alpha_b(z - \xi_b) - uQ_b(z)$ has slope $\alpha_{b,u}$ and root $(\xi_b)_u$. Summing these polynomials gives $z - h - u$, so the weights sum to one and their weighted actions to $h + u$. At either clock the polynomial equals $(T_j - h - u)p_b(T_j)$. This proves the fibre assertion and differentiation gives (3.3). In the perturbation arguments below this gauge polynomial is used algebraically even if it leaves the parameter interval; its slopes stay positive and its roots remain in a fixed slightly larger interval. $\square$

Let $\mathcal{R}$ be the subset of the affine model image on which $\text{rank } P_1 = k$ and $P_2 \neq P_3$. On $\mathcal{R}$, the preceding lemma recovers h , then the two coefficients A, C of $K(z) = zA - C$. Choose orthonormal bases Q_1, Q_2 of the column and row spaces of A , and set

$$A_0 = Q_1^T A Q_2, \quad C_0 = Q_1^T C Q_2, \quad M(z) = zA_0 - C_0. \quad (3.4)$$

For each distinct generalized eigenvalue x , define

$$R_x = \text{Res}_{z=x} M(z)^{-1}, \quad \beta(P) = \sum_x \|R_x\|_{\text{op}}, \quad \eta(P) = (\beta(P) + \gamma(P)^{-1})^{-1}. \quad (3.5)$$

Set $\eta = 0$ off $\mathcal{R}$. The residues are residues of a matrix function, not rank-one factors chosen inside repeated eigenspaces.

Theorem 3.2 (An observable affine singular invariant). *The quantities β and η do not depend on the chosen orthonormal bases or on a latent representation. On the affine model image,*

$$\eta(P) > 0 \iff \vartheta_*(P) > 0 \iff \text{rank } P_1 = k \text{ and } P_2 \neq P_3. \quad (3.6)$$

For every representation of $P \in \mathcal{R}$,

$$\sigma_k(A)^{-1} \leq \beta(P) \leq \frac{k}{\alpha_* \kappa}, \quad \eta(P) \geq c\vartheta_*(P). \quad (3.7)$$

Every competitor in the full closed affine model satisfies

$$d_\infty(\xi, \xi') + |h - h'| \leq C \min\{1, \delta/\eta(P)\}. \quad (3.8)$$

All constants have precisely the fixed-parameter dependencies stated in Section 1; no individual action gap enters them.

On $\mathcal{R}$, β is lower semicontinuous and is continuous on each fixed action-multiplicity stratum. The extension of η by zero is upper semicontinuous on the whole model image, continuous on those strata, and tends to zero at its boundary zero set. The envelope ϑ_* is upper semicontinuous on the whole image.

Proof. For any representation, put $S = Q_1^\top U$ and $T = Q_2^\top V$. These are invertible, and

$$M(z)^{-1} = T^{-\top} \operatorname{diag} \left(\frac{1}{\alpha_b(z - \xi_b)} \right) S^{-1} = \sum_x \frac{R_x}{z - x}, \quad R_x = T^{-\top} \operatorname{diag}(\alpha_b^{-1} \mathbf{1}_{\xi_b=x}) S^{-1}. \quad (3.9)$$

Thus every pole is simple, even when the corresponding action has higher multiplicity. An orthogonal change of bases multiplies each residue on its left and right by orthogonal matrices, preserving its operator norm. The construction otherwise uses only h, A, C . Taking the coefficient at infinity gives $\sum_x R_x = A_0^{-1}$. Each residue norm is at most $1/(\alpha_* \kappa)$, proving (3.7) before its last assertion. Every representation of a rank- k observation has full rank, since all diagonal probabilities are positive. Further, $P_2 = P_3$ is equivalent to $B = 0$. This proves (3.6). The bounds on β imply $\eta \geq c\vartheta$ for every representation, hence $\eta \geq c\vartheta_*$.

To prove the inverse, it suffices to consider $\delta(\beta + \gamma^{-1})$ small. The pole lemma gives $u = h' - h = O(\delta/\gamma)$. Let $Q(z)$ be the affine interpolant of P_1, P_2 , and form the true gauge polynomial $\overline{K}_u(z) = K(z) - uQ(z)$. Its reduced inverse is

$$\overline{M}_u(z)^{-1} = \sum_x \frac{R_x/a_x(u)}{z - x_u}. \quad (3.10)$$

For sufficiently small $|u|$, uniformly in the actions, $1/2 \leq a_x(u) \leq 3/2$ and $|x_u - x| \leq C|u|$. The fractional-linear map $x \mapsto x_u$ has positive derivative close to one on $[L, D]$, so it preserves distinct groups and their multiplicities. Consequently

$$\|\overline{M}_u(z)^{-1}\| \leq \frac{2\beta(P)}{\operatorname{dist}(z, \{x_u\})}.$$

On the other hand, $M'(z) = Q_1^\top K'(z) Q_2$ satisfies $M' = \overline{M}_u + E$, where E is the affine interpolant of

$$(T_j - h')Q_1^\top (P'_j - P_j)Q_2, \quad j = 1, 2.$$

Its coefficient norm is at most $C\delta$, with no channel inverse. The leading inverse of $\overline{M}_u$ has norm at most 2β , so the leading coefficient remains invertible throughout $\overline{M}_u + sE$, $0 \leq s \leq 1$. The same resolvent bound first excludes roots outside a fixed disk and then excludes them outside disks of radius $C\beta\delta$ about the x_u . The contour and multiplicity argument in Lemma 2.4, now with degree k , gives a matching within $C\beta\delta$. The invertible leading coefficient of M' forces both reduced competitor channel factors to be invertible. Its roots are therefore precisely ξ' . Combining this matching with $|x_u - x| \leq C\delta/\gamma$ proves (3.8). Compactness covers larger errors.

For regularity, let $P^{(n)} \rightarrow P$ in $\mathcal{R}$. Locally one can choose continuous orthonormal frames for the rank- k column and row spaces, for example by projecting fixed independent columns and orthonormalizing. The recovered h, A, C , hence the reduced pencils, then converge. Choose disjoint circles enclosing the distinct actions of the limit pencil and no others. The inverse pencils converge uniformly on these circles. Integrating gives

$$\sum_{y \text{ in the cluster of } x} R_y^{(n)} \longrightarrow R_x.$$

The triangle inequality and summation imply $\beta(P) \leq \liminf_n \beta(P^{(n)})$. If the multiplicity pattern is fixed locally, each circle contains one distinct pole, so each residue converges and β is continuous. Since $\gamma > 0$ is continuous here, this gives the assertions for η on $\mathcal{R}$.

At a constant-curve limit, the difference formula in the pole lemma, uniformly for $h \in [L, D]$, gives $\gamma \rightarrow 0$, hence $\eta \rightarrow 0$. At a nonconstant rank-loss limit, h, A still converge and $\sigma_k(A) \rightarrow 0$. Since $\beta \geq \sigma_k(A)^{-1}$, again $\eta \rightarrow 0$. This proves the extension statement. Finally, choose maximizing representatives in (1.7) along a sequence realizing a limsup. A compact subsequence converges to a representation of the limit datum. Continuity of ϑ proves upper semicontinuity of ϑ_* . $\square$

Remark 3.3 (Why semicontinuity is the appropriate statement). Take $k = m_1 = m_2 = 3$, $U = V = I$, $\alpha_b = 1/3$, and actions $(a - t, a + t, b)$ with $a < b$ and small $t \geq 0$. For $t > 0$ the three residue norms equal 3, so $\beta = 9$. At $t = 0$ the residue at a is $3 \operatorname{diag}(1, 1, 0)$, while the residue at b has norm 3; hence $\beta = 6$. The contrast stays positive. Thus η has an upward jump at this partial collision and is not globally continuous on $\mathcal{R}$. This example does not assert discontinuity of ϑ_* . The zero-set statement is a characterization of the certificate, not a claim that every rank-deficient datum loses every possible spectral feature, nor a distance-to-discriminant formula.

Proposition 3.4 (Exact factors and the original affine modulus). *For a full-rank nonconstant affine datum, every exact representation has the same h and spectrum and has full-rank channels. If the spectrum is simple, the channels and weights are unique up to a common permutation. At a repeated action, its total matrix coefficient is unique, but its individual positive rank-one factors need not be. Moreover, if $\xi_k - \xi_1 \geq \Delta > 0$, then $\vartheta \geq c_\Delta \kappa$. The original one-sided estimate with ϑ_* and all its gap-free consequences follow from (3.8).*

Proof. The pole is determined by (3.1). Hence A, C are determined, and $\operatorname{rank} A = k$. In an alternative exact representation $A = U' \operatorname{diag}(\alpha') V'^T$; its rank forces both alternative channels to have full column rank. This justifies applying spectral reconstruction to every point of the exact fibre, not just to a selected full-rank representative.

Choose row and column index sets I, J with A_{IJ} invertible. The spectral projector Π_x of $K(T_1)_{IJ} A_{IJ}^{-1}$ at $T_1 - x$ gives

$$E_x = A_{:J} A_{IJ}^{-1} \Pi_x A_{I:} = \sum_{\xi_b = x} \alpha_b U_{:b} V_{:b}^T.$$

For a simple action, entry sum, row sums and column sums of E_x recover $\alpha_b, U_{:b}, V_{:b}$. At a repeated action only their displayed sum is determined. Finally,

$$\gamma \geq \kappa \left(\sum_b \alpha_b^2 (h - \xi_b)^2 \right)^{1/2} \geq \frac{1}{2} \alpha_* \Delta \kappa.$$

Together with $\vartheta = \kappa \gamma / (\kappa + \gamma)$ this proves the last claim. In particular, on every simple-spectrum exact fibre, $\vartheta_* = \vartheta$. $\square$

4. A DIRECT SPECTRAL ESTIMATOR

The inverse estimates concern the true experiment, but their proof does not require a global search over latent factors. We give an explicit estimator for the multiset alone. Computation of the exact residual confidence region in the next section is a separate problem.

Given matrices $\hat{P}_j$, form $\mathcal{N}_{\hat{P}}$ and $b_{\hat{P}}$ by (1.2). If the former has full column rank, let $\hat{a}$ be its least-squares solution and put $\hat{q}(z) = z^d + \sum_{r < d} \hat{a}_r z^r$. For $d = 1$, replace $-\hat{a}_0$ by its projection onto $[L, D]$; write the resulting polynomial as $z - \hat{h}$. Interpolate the matrices $\hat{q}(T_j) \hat{P}_j$ at the first $d + 1$ clocks to obtain $\widehat{K}(z)$.

Take the leading k left and right singular subspaces of $\hat{P}_1$, with orthonormal bases $\hat{Q}_1, \hat{Q}_2$. Form the reduced polynomial

$$\widehat{M}(z) = \hat{Q}_1^T \widehat{K}(z) \hat{Q}_2.$$

Compute its kd roots, counted with algebraic multiplicity, project the real parts onto $[L, D]$, and sort them. This is the estimator $\hat{\Lambda}$. If the normalization matrix lacks full column rank, a sampled $\hat{q}$ is nonpositive, $\hat{P}_1$ has rank less than k , or the leading coefficient of $\widehat{M}$ is singular, return a fixed multiset in $[L, D]$ instead. This convention defines the estimator everywhere. Numerical root tolerances can be added to the bounds below; the statements themselves refer to exact arithmetic.

Theorem 4.1 (Constructive recovery). *Put $e = \max_j \|\hat{P}_j - P_j\|_F$. At every polynomial truth with $\kappa > 0$ and $\tau(P) > 0$, the estimator just defined satisfies*

$$d_\infty(\hat{\Lambda}, \Lambda) \leq C \min\{1, [e(\kappa^{-1} + \tau(P)^{-1})]^{1/d}\}. \quad (4.1)$$

The within-component simple-root hypothesis of Theorem 1.1 gives the corresponding linear bound. In the affine experiment the stronger observable estimate holds:

$$d_\infty(\hat{\xi}, \xi) \leq C \min\{1, e/\eta(P)\}, \quad P \in \mathcal{R}. \quad (4.2)$$

The procedure uses a least-squares solve, a singular value decomposition, polynomial interpolation, and the roots of a regular matrix polynomial of size k and degree d . It does not optimize over U, V or enumerate a parameter net. No conditioning floor is an input to the estimator.

Proof. The normalization operator and its right-hand side depend linearly on the observed entries. Therefore $\|\mathcal{N}_{\hat{P}} - \mathcal{N}_P\| \leq Ce$, and the residual of the true coefficient vector in the perturbed equation is at most Ce . If $e \leq c\tau$, the least singular value is at least $\tau/2$. Least-squares optimality followed by the triangle inequality gives

$$\|\hat{a} - a\| \leq Ce/\tau. \quad (4.3)$$

Projection of $\hat{h}$ cannot increase its distance from h . For sufficiently small e/τ , all sampled denominator values remain positive.

We record the subspace estimate used in both parts of the proof. Let Q_1, Q_2 span the true column and row spaces of P_1 . The best rank- k approximation property implies

$$\|(I - \hat{Q}_1 \hat{Q}_1^\top)P_1\| \leq 2e, \quad \|P_1(I - \hat{Q}_2 \hat{Q}_2^\top)\| \leq 2e.$$

If $2e/\sigma_k(P_1)$ is sufficiently small, the largest principal angle has sine at most that ratio. In particular both matrices $\hat{Q}_1^\top Q_1$ and $Q_2^\top \hat{Q}_2$ are invertible with inverse norms at most two. This assertion also follows directly by applying the preceding residual inequalities to a singular value decomposition of P_1 .

For the polynomial bound, $\sigma_k(P_1) \geq c\kappa$. Consequently $S = \hat{Q}_1^\top U$ and $T = \hat{Q}_2^\top V$ are invertible, with $\|S^{-1}\| \|T^{-1}\| \leq C/\kappa$, when e/κ is small. At the interpolation nodes, reduction by these inverses yields

$$\begin{aligned} S^{-1} \widehat{M}(T_j) T^{-\top} - D_f(T_j) &= \hat{q}(T_j) S^{-1} \hat{Q}_1^\top (\hat{P}_j - P_j) \hat{Q}_2 T^{-\top} \\ &\quad + (\hat{q}(T_j) - q(T_j)) \text{diag}(p_b(T_j)). \end{aligned}$$

As in (2.2), interpolation bounds the coefficient error by $Ce(\kappa^{-1} + \tau^{-1})$. The leading coefficient is then invertible and Lemma 2.4 applies. The real-part and interval projections cannot increase the distance to any matched true real root. Sorting gives the claimed multiset estimate. All failure conditions are excluded in this small-error regime; bounded range handles its complement.

For the stronger affine assertion, no comparison with κ is needed. The resolvent formula (3.9) gives

$$\sigma_k(P_1)^{-1} = \|(T_1 - h)M(T_1)^{-1}\| \leq C\beta(P).$$

Thus the two estimated subspaces have the preceding inverse bounds as soon as $e\beta$ is small. The true pencil, reduced into these estimated subspaces, has residues

$$(Q_2^\top \hat{Q}_2)^{-1} R_x (\hat{Q}_1^\top Q_1)^{-1},$$

whose total norm is at most 4β . By (4.3) and $\tau \asymp \gamma$, $u = \hat{h} - h$ satisfies $|u| \leq Ce/\gamma$. Apply the exact gauge (3.2) to this true reduced pencil. The difference between the estimated pencil and that gauge polynomial is the affine interpolant of

$$(T_j - \hat{h}) \hat{Q}_1^\top (\hat{P}_j - P_j) \hat{Q}_2, \quad j = 1, 2,$$

and has coefficient norm at most Ce . The resolvent and root-counting argument of Theorem 3.2 now gives root matching error $C\beta e$ about the shifted actions, plus Ce/γ for the shift itself. It applies to complex estimated roots as well. Projection and the bounded-range convention complete (4.2). $\square$

The theorem is an arithmetic and stability statement for this explicit procedure. It is not a bit-complexity theorem uniform in dimension, precision and conditioning. In degree one only a $k \times k$ generalized eigenvalue problem is required. Confidence-region optimization is not being declared tractable on the strength of this point-estimation result.

5. HONEST INFERENCE AND SHARP OBSERVABLE SIGNAL CLASSES

We first record the compact-quotient principle behind residual confidence sets. It applies to the polynomial model as well as its affine specialization.

Lemma 5.1 (Residual inversion). *Let a compact parameter space have a continuous observation map F into a finite product of probability simplices and a continuous target g in a compact metric space. For a true parameter ω , put*

$$\mathbf{m}_\omega(t) = \sup\{d(g(\omega), g(\omega')) : \|F(\omega') - F(\omega)\| \leq t\}.$$

Then $\mathbf{m}_\omega(0) = 0$ exactly when the target is constant on the exact fibre. In that case $\mathbf{m}_\omega(t) \rightarrow 0$ as $t \downarrow 0$. If $e = \|\hat{F} - F(\omega)\|$, a minimum-residual estimate has target error at most $\mathbf{m}_\omega(2e)$. The target image of a residual ball of radius ρ about $\hat{F}$, when nonempty, has diameter at most $2\mathbf{m}_\omega(e + \rho)$ and contains the truth on $e \leq \rho$.

Proof. The zero-radius assertion is the definition. If continuity at zero failed, compactness would give a limiting competitor in the exact fibre with different target. A residual minimizer has residual at most e , because the truth is feasible. The triangle inequality gives observation error at most $2e$. Every point of the residual ball is at observation distance at most $e + \rho$ from the truth, proving the remaining assertions. $\square$

For the affine experiment, take n independent categorical draws at each of the three clocks and put $N = 3n$, $r = m_1 m_2$. Let $\hat{P}_j$ be the empirical matrices. For $0 < \zeta < 1$, define

$$\rho_{N,\zeta} = \sqrt{\frac{r}{2n} \log \frac{6r}{\zeta}}, \quad \mathcal{C} = \{\omega \in \mathcal{K}_1 : \max_j \|P_\omega(T_j) - \hat{P}_j\|_F \leq \rho_{N,\zeta}\}. \quad (5.1)$$

Its target image is $\mathcal{C}_\xi$. If it is empty, report model inconsistency and the empty set. No action is assigned by the confidence procedure in that case. Coordinate extrema of the ordered spectra give simultaneous bands whenever the set is nonempty.

Theorem 5.2 (Risk, honesty, and adaptation). *The set (5.1) has coverage at least $1 - \zeta$ uniformly over the entire closed model, including singular boundaries. The empty set is reported explicitly, and its diameter is defined as zero solely for the expected-diameter loss. For either $\psi = \eta$ or $\psi = \vartheta_*$, and every true datum with $\psi(P) > 0$, the direct estimator of Theorem 4.1 satisfies*

$$\begin{aligned} \mathbb{E}_P d_\infty(\hat{\xi}, \xi)^2 &\leq C \min\{1, (N\psi(P)^2)^{-1}\}, \\ \mathbb{E}_P \text{diam}_\infty \mathcal{C}_\xi &\leq C_\zeta \min\{1, (\sqrt{N}\psi(P))^{-1}\}. \end{aligned} \quad (5.2)$$

On the coverage event the diameter is at most $C \min\{1, \rho_{N,\zeta}/\psi(P)\}$. Neither signal nor either channel factor is supplied to the procedures. At $\psi = 0$ the statements are coverage and bounded-range loss, not shrinking diameter.

Proof. Hoeffding's inequality and a union bound over the $3r$ cell frequencies give

$$\Pr\{e > \rho_{N,\zeta}\} \leq 6r \exp(-2n\rho_{N,\zeta}^2/r) = \zeta.$$

An entrywise error bound t gives a Frobenius bound $\sqrt{r}t$; for binary readouts the factor is 2, not 1. Theorem 3.2 and Lemma 5.1 give the eventwise assertions, and on every outcome with nonempty set give $\text{diam } \mathcal{C}_\xi \leq C \min\{1, (e + \rho_{N,\zeta})/\psi(P)\}$. Further,

$$\mathbb{E}e^2 \leq \sum_{j=1}^3 \mathbb{E}\|\hat{P}_j - P_j\|_F^2 = \frac{1}{n} \sum_{j=1}^3 \left(1 - \sum_{a,b} (P_j)_{ab}^2\right) \leq 3/n.$$

No exceptional outcome is discarded. The empty-set convention preserves the diameter bound. Taking expectations proves its asserted order. Squaring (4.2) and using this second moment proves the risk assertion; $\eta \geq c\vartheta_*$ supplies the other signal. Compactness gives extrema of the residual region. For fixed n the categorical sample space is finite, so choosing minimizers, singular bases and tie conventions causes no measurable selection issue. $\square$

Exact computation of $\mathcal{C}_\xi$ is an information-theoretic construction; it is not implied to be efficient by the preceding point estimator. Finite forward nets with mesh r_0 provide an exhaustive approximation, with residual padding Cr_0 and action padding r_0 , by uniform forward Lipschitz continuity. This may be exponential in dimension and resolution. For degree $d > 1$, Lemma 5.1 and Theorem 1.1 give the analogous polynomial confidence modulus. The following sharp statements concern the affine experiment.

5.1. Two positive least-favourable families. We give all constants in one lemma, so that the parameter class and the testing displacement use the same admissible neighbourhood.

Let $u_0 = m_1^{-1}\mathbf{1}$, $v_0 = m_2^{-1}\mathbf{1}$. Choose unit zero-sum vectors e, f and, when $k > 2$, orthonormal zero-sum vectors $e_3, \dots, e_k$ perpendicular to e and $g_3, \dots, g_k$ perpendicular to f . The dimension assumption $m_i \geq k$ gives these vectors. Fix a small constant $c_0 > 0$ and interior distinct actions whose first pair is $a_0 - b_0, a_0 + b_0$, with $b_0 > 0$, all other actions being above this pair. Set $\alpha_b = 1/k$. For $0 \leq t \leq t_0 < b_0/2$, let $\lambda = 1 - t/b_0$ and put

$$\begin{aligned} \xi_{1,t} &= a_0 - b_0 + t, & \xi_{2,t} &= a_0 + b_0 - t, & \xi_{b,t} &= \xi_b \quad (b \geq 3), \\ U_{t,:1} &= u_0 + \frac{\epsilon_u e}{2\lambda}, & U_{t,:2} &= u_0 - \frac{\epsilon_u e}{2\lambda}, & U_{t,:b} &= u_0 + c_0 e_b \quad (b \geq 3), \\ V_{t,:1} &= v_0 + \frac{\epsilon_v f}{2\lambda}, & V_{t,:2} &= v_0 - \frac{\epsilon_v f}{2\lambda}, & V_{t,:b} &= v_0 + c_0 g_b \quad (b \geq 3). \end{aligned} \tag{5.3}$$

The pole h is fixed along this family.

The second family fixes positive full-rank channels and positive weights. Choose distinct real z_b with $\sum_b \alpha_b z_b = 0$ and put, for an interior a ,

$$\xi_b(t) = a + t + sz_b, \quad 0 \leq t \leq t_0, \quad B = -sU \text{diag}(\alpha_b z_b) V^\top. \tag{5.4}$$

Here $h_t = a + t$, while A, B do not depend on t .

Lemma 5.3 (Uniform family constants). *There exist fixed $t_0 > 0$, $\epsilon_* > 0$, $s_0 > 0$, a cell floor $p_* > 0$, and positive constants c_1, C_1, C_2 , depending only on the fixed experiment and the fixed choices just made, with the following properties. Both families are admissible throughout these ranges, all joint cell probabilities are at least p_* for every $T \in [T_1, T_3]$, and $d_\infty(\xi(t), \xi(0)) = t$. For either signal $\psi \in \{\eta, \vartheta_*\}$ and positive $w = \epsilon_u \epsilon_v$ in (5.3), or $w = s$ in (5.4), uniformly on $[0, t_0]$,*

$$c_1 w \leq \psi(P_t) \leq C_1 w, \quad \sup_{T \in [T_1, T_3]} \|P_t(T) - P_0(T)\|_F \leq C_2 w t. \tag{5.5}$$

In the product family both observed marginals are exactly fixed, and

$$P_t(T) - P_0(T) = \frac{T - a_0}{2k(T - h)} \epsilon_u \epsilon_v (\lambda^{-2} - 1) e f^\top, \quad (5.6)$$

$$\kappa(\omega_t) = \frac{\epsilon_u \epsilon_v}{2\lambda^2}, \quad \sigma_k(A_t) = \kappa(\omega_t)/k. \quad (5.7)$$

In the collapse family,

$$P_t(T) - P_0(T) = \frac{tB}{(T - a - t)(T - a)}. \quad (5.8)$$

If $w = 0$, the respective family has exactly the same laws for all t and all clocks in the interval, despite its changing action multiset.

Proof. Choose c_0, ϵ_* small enough that every channel entry in the product family is at least half the corresponding uniform entry. Shrink t_0 and s_0 to keep every action in the declared interval and to preserve order. The collapse family also has a fixed positive cell floor, since its channels are strictly positive. These choices precede all later choices of N, w and the testing displacement.

In the first two terms of the product family the unnormalized sum is

$$(T - a_0 + \lambda b_0)(u_0 + \epsilon_u e / (2\lambda))(v_0 + \epsilon_v f / (2\lambda))^\top \\ + (T - a_0 - \lambda b_0)(u_0 - \epsilon_u e / (2\lambda))(v_0 - \epsilon_v f / (2\lambda))^\top.$$

Its constant part is $2(T - a_0)u_0v_0^\top$, its linear parts are $b_0\epsilon_u\epsilon_vv_0^\top + b_0\epsilon_vu_0f^\top$, and its interaction part is $(T - a_0)\epsilon_u\epsilon_v e f^\top / (2\lambda^2)$. Division by $k(T - h)$ proves (5.6). Zero sums of e, f prove exact marginal cancellation.

The latent vector $(1, -1, 0, \dots, 0)/\sqrt{2}$ is a right singular vector of U_t with singular value $\epsilon_u/(\sqrt{2}\lambda)$. Its orthogonal complement maps, independently of ϵ_u , into the independent directions $u_0, e_3, \dots, e_k$, all perpendicular to e . Its least singular value is a fixed positive number. Choosing ϵ_* smaller if necessary gives the stated smallest singular value, and the same proof applies to V_t . In $A_t = U_t V_t^\top / k$ the pair direction maps f to $(\kappa/k)e$; the complementary rank- $(k - 1)$ map is fixed and has its nonzero singular values bounded below. This proves (5.7).

Taking the marginals of B gives components of magnitude $b_0\epsilon_u/k$ and $b_0\epsilon_v/k$ in the e and f directions. Thus $\gamma \geq c \max(\epsilon_u, \epsilon_v)$ and $\vartheta \asymp \epsilon_u \epsilon_v$. The actions are simple and both channels have full rank when the two contrasts are positive. Proposition 3.4, including its rank argument for every alternative exact representation, gives $\vartheta_* = \vartheta$. Further, (3.7) and (5.7) imply $\beta \asymp \kappa^{-1}$; hence $\eta \asymp \epsilon_u \epsilon_v$. Since λ stays in a fixed compact subinterval of $(0, 1]$, $\lambda^{-2} - 1 \asymp t$, uniformly on the chosen interval.

For the collapse family, subtraction of $A + B/(T - a - t)$ gives (5.8). The fixed channel ranks imply $\gamma \asymp s$, while κ stays bounded below. Simple-spectrum uniqueness gives $\vartheta_* = \vartheta \asymp s$ for $s > 0$. The inequalities $c\vartheta_* \leq \eta \leq \gamma$ give $\eta \asymp s$. The denominators in both exact identities have fixed positive bounds, which gives (5.5) with one common set of constants after taking minima and maxima. Both exact identities also prove the assertions at $w = 0$. $\square$

5.2. Adaptive experiments and the lower bounds. An adaptive experiment consists of parameter-independent probability kernels $\pi_i(dT \mid \mathcal{H}_{i-1})$ on $[T_1, T_3]$. Here $\mathcal{H}_{i-1}$ contains all preceding clocks and categorical observations, and any external randomization independent of the parameter. At step i sample the clock with this kernel and then one categorical observation with law $P_\omega(T_i)$. Deterministic predictable clocks and fixed designs are included. The complete transcript, including clocks, is available to the estimator.

Theorem 5.4 (Sharp classes and honest-diameter lower bounds). *For either $\psi = \eta$ or $\psi = \vartheta_*$, there exists $s_* > 0$ such that for $0 < s \leq s_*$ and $N = 3n$,*

$$\inf_{\hat{\xi}} \sup_{\omega: \psi(\mathcal{P}(\omega)) \geq s} \mathbb{E}_\omega d_\infty(\hat{\xi}, \xi)^2 \asymp \min\{1, (Ns^2)^{-1}\}. \quad (5.9)$$

The direct estimator at the balanced three-clock design gives the upper bound. The lower bound remains valid when the infimum includes every adaptive experiment just defined.

If a random spectrum set J is honest at level $1 - \zeta$ over $\mathcal{K}_1$, with $0 < \zeta < 1/4$, then at the base point of either positive family,

$$\mathbb{E}_0 \text{diam}_\infty J \geq c_\zeta \min\{1, (\sqrt{N}\psi(P_0))^{-1}\}. \quad (5.10)$$

This is true under any such clock strategy. The exact singular limits of these families have nonvanishing risk and honest-diameter lower bounds. Equal channel contrasts ϵ give the transition $N\epsilon^4 \asymp 1$; full-spectrum spread s with fixed channels gives $Ns^2 \asymp 1$.

Proof. The upper bound follows from Theorem 5.2. Lemma 5.3 and its cell floor give, uniformly in the clock,

$$\text{KL}(P_t(T), P_0(T)) \leq \sum_{a,b} \frac{((P_t)_{ab} - (P_0)_{ab})^2}{(P_0)_{ab}} \leq Cw^2t^2.$$

The conditional clock kernel is the same under both parameters. The relative-entropy chain rule therefore gives

$$\text{KL}(\mathbb{P}_t^{(N)}, \mathbb{P}_0^{(N)}) = \sum_{i=1}^N \mathbb{E}_t \text{KL}(P_t(T_i), P_0(T_i)) \leq CNw^2t^2.$$

This equality includes the external randomization and the entire transcript, so no fixed-design assumption is hidden in the bound. Take $t = c \min\{t_0, (\sqrt{N}w)^{-1}\}$ with a sufficiently small fixed c . Pinsker's inequality makes the total variation at most $1/2$. The spectra are distance t apart. Turning an estimate into the nearer of the two spectra gives maximal squared risk at least $t^2/16$.

Here is the class bookkeeping. Fix once and for all the constants in Lemma 5.3, choose $M \geq 1/c_1$, and set $w = Ms$, $\epsilon_u = \epsilon_v = \sqrt{Ms}$ in the product family. Take $s_* \leq \epsilon_*^2/M$. Then every $t \in [0, t_0]$, including the testing displacement just chosen, has positive cell floor p_* , admissible actions and weights, and $\psi(P_t) \geq c_1Ms \geq s$. All constants in the entropy estimate remain unchanged. This proves the lower bound on the exact class in (5.9); the collapse family gives the other attained singular order.

For an honest set, transfer coverage at the alternative to the base law. The union bound gives

$$\Pr_0\{\xi(0), \xi(t) \in J\} \geq 1 - 2\zeta - \text{TV}(\mathbb{P}_t^{(N)}, \mathbb{P}_0^{(N)}).$$

Decreasing c depending on ζ leaves a positive fixed lower bound. On this event the diameter is at least t . The comparison $\psi(P_0) \asymp w$ proves (5.10). At $w = 0$ use a fixed nonzero t ; all conditional laws agree at every clock, so the transcript laws are identical for every policy. The same testing and coverage arguments give the stated nonvanishing bounds. $\square$

The two paths identify different losses of the observation map. The first removes both first-order marginal changes and leaves a tensor product of weak channel directions. The second keeps the channels fixed and removes the whole projective contrast. Neither construction charges a small gap between two otherwise visible actions. The results are class-sharp and pathwise sharp as stated, rather than a complete classification of all directional moduli at intersections of singular strata.

6. RELATION TO THE OTHER OBSERVATION LAWS

In degree one, normalizing raw matrices $c(T)U \operatorname{diag}(a_b(T - \xi_b))V^\top$, with an unknown positive scalar detector $c(T)$, gives (1.1) with $\alpha_b = a_b / \sum_c a_c$. In particular the logarithmic action law is unchanged:

$$\log \frac{p_b(T)}{p_c(T)} = \log \frac{\alpha_b}{\alpha_c} + \log(T - \xi_b) - \log(T - \xi_c).$$

For $k = m_1 = m_2 = 2$, stochastic columns imply

$$\frac{|\det U|}{\sqrt{2}} \leq \sigma_2(U) \leq |\det U|, \quad \frac{|\det V|}{\sqrt{2}} \leq \sigma_2(V) \leq |\det V|.$$

Indeed the largest singular value lies between 1 and $\sqrt{2}$, and the product of the two singular values is the absolute determinant. On an action class with $\xi_2 - \xi_1 \geq \Delta > 0$, Proposition 3.4 therefore gives the single inverse power of $|\det U \det V|$. The identity $\alpha_1 = (\xi_2 - h)/(\xi_2 - \xi_1)$ also recovers the relative throughput with a constant depending on this fixed gap.

The shared two-clock spatial experiment has a different nuisance space: the clock factors are common across sites. Spatial calibration can replace a third clock there. Its exact identification and one-clock geometric alternatives, and its local correctly-labelled calibration estimate, are separate from the pointwise projective experiment treated here. The local estimate is not a global bound through collisions of spatial profiles.

The polynomial, rational and compact-surface detector quotients in the historical derivations retain their respective secant and tangent tests, divisor counts, and branched-clock forms of the logarithmic law. Theorem 1.1 concerns the specific simultaneous polynomial numerator and its scalar normalization. Its normalization-rank criterion and sharp worst-case $2d + 1$ count do not declare every ambient meromorphic pole-budget bound optimal. In particular, common-factor cancellation in (1.3) is a property of this projective observation, not an identification of all those detector quotients with one another.

Finite-record geometric certificates, component coverings, return-lift constructions, and raw-acquisition laws remain logically separate. Where the recovered actions determine boundary-distance data, metric rigidity is supplied by the stated classical geometric rigidity hypotheses in those derivations. The present results concern the upstream observation inverse. All proofs of the principal results above are contained in this article and require no geometric appendix; the other regimes are retained as an independent archive rather than added as hypotheses to this theorem.

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